QUESTION ANALYSIS

EEE337

Power System II

Question Analysis

Sessions 2021–22 to 2020–21 *2 papers, 51 questions, 96 formulas*

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1. At a glance:

This analysis covers **2 papers** (2021-22, 2020-21), with 51 questions and 325 marks offered. It shows what past papers asked. It does not predict the next paper, and with so few papers a pattern can easily be chance.

The five topics with the most marks:

Topic	Marks	Share	Questions	Top subtopics (marks)
Underground cables	85	26 %	11	Dielectric stress (43); Grading (20); Capacitance (10)
Line inductance	66	20 %	11	Mutual inductance (25); Composite conductors (18); Transposition (12)
Overhead insulators	48	15 %	7	String efficiency (41); Failure (7)
Power system stability	43	13 %	5	Equal area criterion (23); Transient EMF (20)
Line capacitance	27	8 %	5	Unsymmetrical spacing (25); Effect of earth (2)

In the topic pages, a formula name in bold is a key formula, the number after a subtopic is its marks offered, and a blue symbol in an equation links to its meaning in the symbol list, where the arrow leads back to the equation.

2. Heat map:

Marks offered, by topic and paper, in book order. A darker cell means more marks. Click a topic to open its page.

Topic and subtopic	2021-22	2020-21	Term tests	All
Line inductance	**26**	**30**	**10**	**66**
Mutual inductance	**10**	**10**	**5**	25
Composite conductors	**8**	**10**	·	18
Transposition	**8**	**4**	·	12
Bundled conductors	·	**4**	**5**	9
Basics	·	**1**	·	1
Skin effect	·	**1**	·	1
Line capacitance	**17**	**10**	·	**27**
Unsymmetrical spacing	**15**	**10**	·	25
Effect of earth	**2**	·	·	2
Corona	**2**	**6**	·	**8**
Critical disruptive voltage	**2**	**5**	·	7
Corona effect	·	**1**	·	1
Overhead insulators	**28**	**15**	**5**	**48**
String efficiency	**26**	**10**	**5**	41
Failure	**2**	**5**	·	7
Mechanical design	**7**	**10**	·	**17**
Sag	**2**	**10**	·	12
Supports	**5**	·	·	5
Underground cables	**40**	**40**	**5**	**85**
Dielectric stress	**8**	**30**	**5**	43
Grading	**20**	·	·	20
Capacitance	·	**10**	·	10
Construction	**5**	·	·	5
Insulation resistance	**5**	·	·	5

Topic and subtopic	2021-22	2020-21	Term tests	All
Economical size	**2**	·	·	2
Cable faults	**5**	**10**	**5**	**20**
Loop tests	**5**	**10**	**5**	20
Power system stability	**18**	**25**	·	**43**
Equal area criterion	**8**	**15**	·	23
Transient EMF	**10**	**10**	·	20
HVDC and FACTS	**7**	**4**	·	**11**
HVDC	**7**	·	·	7
FACTS	·	**4**	·	4

3. Priority list:

Must know: Underground cables (85), Line inductance (66), Overhead insulators (48), Power system stability (43), Line capacitance (27), Cable faults (20), Mechanical design (17), HVDC and FACTS (11), Corona (8)

Likely: none

Less often: none

The rule: Must know means asked in every paper. Likely means asked in at least half the papers. Less often means asked in fewer than half. Numbers in brackets are marks offered, and each list runs from most marks to fewest.

4. Topic pages:

4.1. Line inductance:

66 marks, 11 questions, in 2 of 2 papers. Book: Mehta Ch. 9; Grainger Ch. 4.

Most often numerical (5 of 11). Subtopics: Basics; Bundled conductors; Mutual inductance; Transposition; Composite conductors; Skin effect.

Formulas:

Subtopic	Formula	Equation	Used in
Basics (1)	**Inductance**	$L = \frac{\psi}{I} = \frac{N\phi}{I}$	20-21 1(a)
	Induced voltage	$e = L\frac{di}{dt}$	20-21 1(a)
Bundled conductors (9)	Internal flux linkage	$\psi_{int} = \frac{\mu_0 I}{8\pi}$	21-22 1(c), 20-21 1(b)
	External flux linkage	$\psi_{12} = \frac{\mu_0 I}{2\pi}\ln\frac{D_2}{D_1} = 2\times10^{-7}\, I \ln\frac{D_2}{D_1}$	20-21 1(b)
	GMR of a solid conductor	$D_s = r' = r\, e^{-1/4} = 0.7788\, r$	20-21 1(b)
	Transposed 3-phase line	$L = 2\times10^{-7}\ln\frac{D_{eq}}{D_s}$ H/m	20-21 1(b)
	Equivalent spacing	$D_{eq} = \sqrt[3]{D_{ab}D_{bc}D_{ca}}$	20-21 1(b)

Subtopic	Formula	Equation	Used in
	Two-conductor bundle	$D_s^b = \sqrt{D_s\, d}$	20-21 1(b)
	Inductive reactance	$X_L = 2\pi f L$	21-22 7(a), 20-21 1(b)
	Two-conductor bundle	$D_s = \sqrt{r'\, d}$	21-22 7(a)
Mutual inductance (25)	**Mutual inductance**	$M = 2\times 10^{-7} \ln \dfrac{D_{P_1T_2}\, D_{P_2T_1}}{D_{P_1T_1}\, D_{P_2T_2}}$ H/m	21-22 1(d), 20-21 1(d)
	Induced voltage	$E = 2\pi f M I$	21-22 1(d), 20-21 1(d)
	Symmetrical arrangement	$M = 4\times 10^{-7} \ln \dfrac{D_2}{D_1}$ H/m	21-22 1(d)
Transposition (12)	Flux linkage in a group	$\psi_a = 2\times 10^{-7}\left[I_a \ln \dfrac{1}{D_s} + I_b \ln \dfrac{1}{D_{ab}} + I_c \ln \dfrac{1}{D_{ac}}\right]$	20-21 2(b)
	Inductance	$L = \dfrac{\psi}{I}$	21-22 1(c)
	External flux linkage	$\psi_{12} = \dfrac{\mu_0 I}{2\pi} \ln \dfrac{D_2}{D_1}$	21-22 1(c)
	GMR of a solid conductor	$r' = r\, e^{-1/4} = 0.7788\, r$	21-22 1(c)
	Symmetrical 3-phase line	$L = 2\times 10^{-7} \ln \dfrac{D}{r'}$ H/m	21-22 1(c)
	Unsymmetrical transposed line	$L = 2\times 10^{-7} \ln \dfrac{D_{eq}}{r'} = 0.2 \ln \dfrac{D_{eq}}{r'}$ mH/km	21-22 1(c)
	Equivalent spacing	$D_{eq} = \sqrt[3]{D_{12} D_{23} D_{31}}$	21-22 1(c)
Composite conductors (18)	**Composite conductor**	$L_X = 2\times 10^{-7} \ln \dfrac{D_m}{D_{sX}}$ H/m	21-22 4(c), 20-21 2(c)
	Mutual GMD	$D_m = \sqrt[mn]{(D_{aa'} D_{ab'} \cdots D_{am}) \cdots (D_{na'} \cdots D_{nm})}$	21-22 4(c), 20-21 2(c)
	Self GMD	$D_{sX} = \sqrt[n^2]{(D_{aa} D_{ab} \cdots D_{an}) \cdots (D_{na} \cdots D_{nn})}$	21-22 4(c), 20-21 2(c)
	Strand inductance	$L_X = \dfrac{L_a + L_b + \cdots + L_n}{n^2}$	20-21 2(c)
	Composite line loop	$L = 2\times 10^{-7} \ln \dfrac{D_m^2}{D_{sX} D_{sY}}$ H/m	21-22 4(c), 20-21 2(c)
Skin effect (1)	Skin depth	$\delta = \dfrac{1}{\sqrt{\pi f \mu \sigma}}$	20-21 3(a)

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Symbols:

L = loop inductance of the line (H/m) ↩

ψ = flux linkage (Wb-T/m) ↩

N = number of turns ↩

ϕ = flux (Wb) ↩

I = current (A) ↩

e = self-induced voltage (V) ↩

i = instantaneous current (A)

t = time (s)

ψ_{int} = flux linkage inside the conductor (Wb-T/m) ↩

μ_0 = permeability of free space ($4\pi \times 10^{-7}$ H/m) ↩

ψ_{12} = flux linkage between two points outside the conductor (Wb-T/m) ↩

D_1, D_2 = distances of the two points from the conductor (m); near and far distances (m) ↩

D_s = geometric mean radius, GMR (m); GMR of the bundle (m) ↩

r = conductor radius (m) ↩

D_{eq} = equivalent spacing (m) ↩

D_{bc}, D_{ca} = the other two spacings (m) ↩

D_s^b = GMR of the bundle (m) ↩

d = spacing of the sub-conductors (m) ↩

X_L = inductive reactance (Ω/m) ↩

f = frequency (Hz) ↩

M = mutual inductance (H/m) ↩

$D_{P_1T_1}$, $D_{P_1T_2}$, $D_{P_2T_1}$, $D_{P_2T_2}$ = distance from power wire i to telephone wire j, $D_{P_iT_j}$ (m) ↩

E = voltage in the telephone line (V per m) ↩

ψ_a = flux linkage of conductor a (Wb-T/m) ↩

I_a, I_b, I_c = currents of the three conductors (A) ↩

D_{ab}, D_{ac} = distances from a to b and to c (m) ↩

r' = geometric mean radius (m) ↩

D = equal spacing of the phases (m) ↩

D_{12}, D_{23}, D_{31} = the three phase spacings (m) ↩

L_X = inductance of conductor X (H/m) ↩

D_m = mutual GMD between conductors X and Y (m) ↩

D_{sX} = self GMD of X (m) ↩

$D_{aa'}$, $D_{ab'}$, D_{am}, $D_{na'}$, D_{nm} = distances from a strand of X to a strand of Y (m) ↩

n, m = strands in X and Y ↩

D_{ab}, D_{an}, D_{na}, D_{nn} = distances between strands of X (m) ↩

L_a, L_b, L_n = inductances of the strands (H/m) ↩

D_{sY} = self GMD of Y (m) ↩

δ = skin depth (m) ↩

μ = permeability (H/m) ↩

σ = conductivity (S/m) ↩

Questions:

No.	Paper	Marks	Type	Question
Basics				
1	20-21 1(a)	1	theory	Define Inductance with the required expression.
Bundled conductors				
2	21-22 TT1 7(a)	5	numerical	A 500 kV line has a bundling arrangement of 2 conductors per phase as shown in the Figure 1. Compute the inductive reactance per phase of this line at 50 Hz. Each conductor carries 50% of the phase current.
3	20-21 1(b)	4	numerical	Each conductor of the bundled-conductor line shown in Fig.1 is ACSR, 1,272,000-cmil Pheasant. Find the inductive reactance in ohms per km (and per mile) per phase for d=45cm.
Mutual inductance				
4	21-22 1(d)	10	numerical	A single-phase 50 Hz power line is supported on a horizontal cross-arm. The spacing between the conductors is 3 m. A telephone line is supported symmetrically below the power line as shown in Fig. 1. Find the mutual inductance between the two circuits and the voltage induced per kilometer in the telephone line if the current in the power line is 100 A. Assume the telephone line current to be zero.
5	21-22 TT3 9(b)	5	numerical repeats 20-21 1(d)	A single phase 50Hz power line is supported on a horizontal cross-arm. The spacing between conductors is 2.5m. A telephone line is also supported on a horizontal cross-arm in the same horizontal plane as the power line. The conductors of the telephone line are of solid copper spaced 0.6m between centers. The distance between the nearest conductors of the two lines is 20m. Find the mutual inductance between the circuits and the voltage per km induced in the telephone line for 150A current flowing over the power line.

No.	Paper	Marks	Type	Question
6	20-21 1(d)	10	numerical	A single phase 50Hz power line is supported on a horizontal cross-arm. The spacing between conductors is 2.5m. A telephone line is also supported on a horizontal cross-arm in the same horizontal plane as the power line. The conductors of the telephone line are of solid copper spaced 0.6m between centers. The distance between the nearest conductors of the two lines is 20m. Find the mutual inductance between the circuits and the voltage per km induced in the telephone line for 150A current flowing over the power line.
Transposition				
7	21-22 1(c)	8	derivation	Define transpose transmission line. Derive the expression for the inductance of three phase transmission lines with unsymmetrical spacing.
8	20-21 2(b)	4	theory	Define Transposed Transmission line. Explain the reason behind Transpose.
Composite conductors				
9	21-22 4(c)	8	derivation repeats 20-21 2(c)	Derive the expression of inductance of composite conductor lines.
10	20-21 2(c)	10	derivation	Derive the expression of Inductance of composite conductor lines.
Skin effect				
11	20-21 3(a)	1	theory	Define Skin Effect.

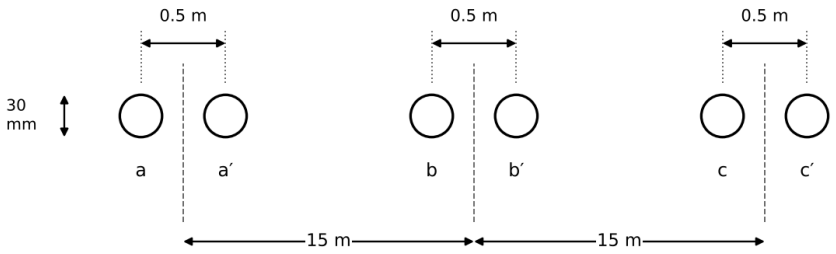


Figure 1: 21-22 TT1 7(a): Figure 1 of Term Test 1 (redrawn)

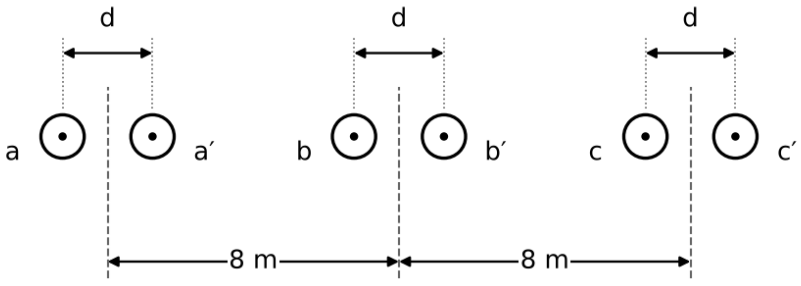


Figure 2: 20-21 1(b): Fig. 1 of the paper (redrawn): figure for Q-1(b)

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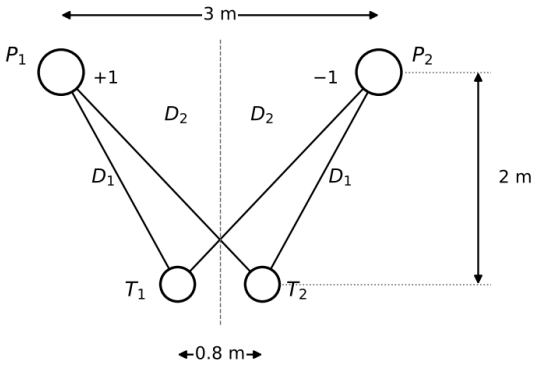


Figure 3: 21-22 1(d): Fig. 1 of the paper (redrawn): figure for Q-1(d)

4.2. Line capacitance:

27 marks, 5 questions, in 2 of 2 papers. Book: Mehta Ch. 9; Grainger Ch. 5.

Most often derivation (3 of 5). Subtopics: Unsymmetrical spacing; Effect of earth.

Formulas:

Subtopic	Formula	Equation	Used in
Unsymmetrical spacing (25)	Capacitance	$C = \frac{q}{V}$	20-21 1(c)
	Field of a line charge	$E = \frac{q}{2\pi\varepsilon_0\, x}$	20-21 1(c)
	Potential difference	$V_{12} = \frac{q}{2\pi\varepsilon_0} \ln \frac{D_2}{D_1}$	21-22 5(d), 20-21 1(c)
	Balanced voltages	$V_{ab} + V_{ac} = 3V_{an}$	20-21 1(c)
	Capacitance to neutral	$C_n = \frac{2\pi\varepsilon_0}{\ln(D_{eq}/r)} = \frac{0.0556}{\ln(D_{eq}/r)}\ \mu\text{F/km}$	20-21 1(c)
	Line charging current	$I_c = 2\pi f\, C_n\, V_{ph}$	21-22 5(b), 20-21 1(c)
	Field of a line charge	$g = \frac{q}{2\pi\varepsilon_0\, x}$	21-22 5(d)
	Capacitance to neutral	$C_n = \frac{2\pi\varepsilon_0}{\ln(D_{eq}/r)}\ \text{F/m}$	21-22 5(d)
Effect of earth (2)	Single-phase line	$C = \frac{\pi\varepsilon_0}{\ln(d/r)}\ \text{F/m}$	21-22 5(a)
	Effect of the earth	$C = \frac{\pi\varepsilon_0}{\ln\left(\frac{d}{r} \cdot \frac{2h}{\sqrt{4h^2 + d^2}}\right)}$	21-22 5(a)

Symbols:

C = capacitance between the two wires (F/m) ↩

q = charge (C/m) ↩

V = voltage (V) ↩

E = electric field (V/m) ↩

ε_0 = permittivity of free space (8.854×10^{-12} F/m) ↩

x = distance from the conductor (m) ↩

V_{12} = voltage between the two points (V) ↩

D_1, D_2 = distances of the two points from the conductor (m) ↩

V_{ab}, V_{ac} = line voltages (V) ↩

V_{an} = phase voltage (V) ↩

C_n = capacitance to neutral per phase (F/m) ↩

D_{eq} = equivalent spacing (m) ↩

r = outside radius of the conductor (m) ↩

I_c = charging current (A) ↩

f = frequency (Hz) ↩

V_{ph} = phase voltage (V) ↩

g = electric field (V/m) ↩

d = spacing of the two wires (m) ↩

h = height above the earth (m) ↩

Questions:

No.	Paper	Marks	Type	Question
Unsymmetrical spacing				
1	21-22 5(b)	5	numerical	A three phase 50 Hz transmission line has flat horizontal spacing with 3.5 m between adjacent conductors. The conductors are No. 2/0 hard/drawn seven strand copper (outside conductor diameter= 1.05 cm). The voltage of the line is 110 kV. Find the capacitance to neutral and the charging current per kilometre of line.
2	21-22 5(d)	10	derivation repeats 20-21 1(c)	Derive the expression of capacitance of a three phase line with unsymmetrical spacing.
3	21-22 TT1 7(b)	0	derivation repeats 21-22 5(d)	Derive the expression for the capacitance of a three-phase line with unsymmetrical spacing.
4	20-21 1(c)	10	derivation	Derive the expression of Capacitance of a three phase line with unsymmetrical spacing.
Effect of earth				
5	21-22 5(a)	2	theory	How Earth affect on transmission line capacitance?

4.3. Corona:

8 marks, 3 questions, in 2 of 2 papers. Book: Mehta Ch. 8.

Most often theory (3 of 3). Subtopics: Corona effect; Critical disruptive voltage.

Formulas:

Subtopic	Formula	Equation	Used in
Corona effect (1)	Field near the conductor	$g = \dfrac{V}{x \ln(d/r)}$	21-22 1(a), 20-21 2(a)
	Strength of air	$g_0 = 30$ kV/cm peak $= 21.2$ kV/cm r.m.s.	21-22 1(a), 20-21 2(a)
Critical disruptive voltage (7)	Surface field	$g_{max} = \dfrac{V}{r \ln(d/r)}$	21-22 1(a), 20-21 4(a)
	Critical disruptive voltage	$V_c = m_0\, g_0\, \delta\, r \ln \dfrac{d}{r}$ kV	21-22 1(a), 20-21 4(a)
	Air density factor	$\delta = \dfrac{3.92\, b}{273 + t}$	21-22 1(a), 20-21 4(a)
	Visual critical voltage	$V_v = m_v\, g_0\, \delta\, r \left(1 + \dfrac{0.3}{\sqrt{\delta r}}\right) \ln \dfrac{d}{r}$	20-21 4(a)

Symbols:

g = potential gradient (kV/cm) ↩

V = phase-to-neutral voltage (kV) ↩

x = distance from the conductor centre (cm) ↩

d = spacing (cm) ↩

r = conductor radius (cm) ↩

g_0 = dielectric strength of air; breakdown strength of air ↩

g_{max} = largest gradient, at the conductor surface (kV/cm) ↩

V_c = critical disruptive voltage (kV) ↩

m_0 = irregularity factor (1 smooth, 0.80–0.87 stranded) ↩

δ = air density factor ↩

b = pressure (cm of Hg) ↩

t = temperature (°C) ↩

V_v = visual critical voltage (kV) ↩

m_v = irregularity factor for visual corona ↩

Questions:

No.	Paper	Marks	Type	Question
Corona effect				
1	20-21 2(a)	1	theory	Define Corona Effect.
Critical disruptive voltage				
2	21-22 1(a)	2	theory repeats 20-21 4(a)	Define critical disruptive voltage. Write down it's significance.
3	20-21 4(a)	5	theory	Define critical disruptive voltage. Write down its significance.

4.4. Overhead insulators:

48 marks, 7 questions, in 2 of 2 papers. Book: Mehta Ch. 8.

Most often theory (3 of 7). Subtopics: String efficiency; Failure.

Formulas:

Subtopic	Formula	Equation	Used in
String efficiency (41)	**Capacitance ratio**	$K = \dfrac{C_1}{C}$	21-22 3(c), 20-21 4(b)
	Unit voltages	$V_2 = (1+K)V_1,\ V_3 = (1+3K+K^2)V_1$	20-21 4(b)
	General unit	$V_{k+1} = V_k + K\,(V_1 + V_2 + \cdots + V_k)$	21-22 8(a), 20-21 4(b)
	String efficiency	$\eta = \dfrac{V}{n\,V_n} \times 100\ \%$	21-22 3(d), 20-21 4(b)
	Three-unit string	$\eta = \dfrac{3+4K+K^2}{3(1+3K+K^2)}$	21-22 3(d), 20-21 4(b)
	Capacitance grading	$C_2 = C_1(1+K),\ C_3 = C_1(1+3K)$	20-21 4(b)
	Unit voltages	$V_2 = V_1(1+K),\ V_3 = V_2 + K(V_1+V_2)$	21-22 3(c)
	String voltage	$V = V_1 + V_2 + \cdots + V_n$	21-22 3(c)
	K from measurements	$K = \dfrac{V_2}{V_1} - 1$	21-22 5(c)
Failure (7)	Safety factor of an insulator	$\text{SF} = \dfrac{\text{puncture strength}}{\text{flash-over voltage}}$	21-22 3(a), 20-21 5(a)

Symbols:

K = ratio of shunt to self capacitance ↩

C_1 = shunt capacitance to the tower (F); shunt capacitance to earth (F)

C = self capacitance of a unit (F) ↩

V_1, V_2, V_3 = voltages across units 1, 2, 3 from the tower (V); voltages across units 1, 2, 3 from the cross-arm (V) ↩

V_k, V_{k+1} = voltages across units k and $k+1$ (V) ↩

η = string efficiency ↩

V = phase voltage across the string (V) ↩

n = number of units ↩

V_n = voltage of the unit nearest the line (V) ↩

C_1, C_2, C_3 = capacitances of units 1, 2, 3 for equal voltages (F) ↩

Questions:

No.	Paper	Marks	Type	Question
String efficiency				
1	21-22 3(c)	8	numerical	In a 33 kV overhead line, there are three units in the string of insulators. If the capacitance between each insulator pin and earth is 11% of self capacitance of each insulator, find (i) the distribution of voltage over 3 insulators and (ii) String efficiency.
2	21-22 3(d)	10	derivation	Derive the mathematical expression of String efficiency with proper diagram. What will happen if: (i) $K = 0.2$ ($K = \frac{shunt\ capacitance}{self\ capacitance}$) (ii) $K = 0.8$ (iii) the number of disc increases in the string.
3	21-22 5(c)	8	numerical	A 3-phase transmission line is being supported by three-disc insulators. The potentials across top unit (near to the tower) and middle unit are 8 kV and 11 kV, respectively. (i) Draw the equivalent circuit (ii) Calculate the ratio of capacitance between pin and earth to the self-capacitance of each unit (iii) Calculate the line voltage and (iv) Calculate string efficiency.
4	21-22 TT2 8(a)	5	numerical	A string of 5 insulators is connected across a 100 kV line. If the capacitance of each disc to earth is 0.1 of the capacitance of the insulator, calculate (i) the distribution of voltage on the insulator discs and (ii) the string efficiency.
5	20-21 4(b)	10	theory	What is String Efficiency? Discuss the methods of improving String Efficiency.
Failure				
6	21-22 3(a)	2	theory	How electrical breakdown happen in a pin-type insulator?
7	20-21 5(a)	5	theory	What are the causes of insulation failure?

4.5. Mechanical design:

17 marks, 3 questions, in 2 of 2 papers. Book: Mehta Ch. 8.

Most often theory (2 of 3). Subtopics: Sag; Supports.

Formulas:

Subtopic	Formula	Equation	Used in
Sag (12)	**Working tension**	$T = \frac{\text{ultimate strength}}{\text{safety factor}}$	20-21 5(c)
	Sag, level supports	$S = \frac{w\,l^2}{8\,T}$	21-22 4(a), 20-21 5(c)
	Conductor curve	$y = \frac{w\,x^2}{2\,T}$	20-21 5(c)
	Height of supports	$h = \text{ground clearance} + S$	20-21 5(c)

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Symbols:

T = working tension (kg-wt or N) ↩ l = span (m) ↩ x = distance from the lowest point (m) ↩

S = sag (m) ↩ y = height above the lowest point (m) ↩ h = height of the supports (m) ↩

w = weight per metre (N/m) ↩

Questions:

No.	Paper	Marks	Type	Question
Sag				
1	21-22 4(a)	2	theory	What is Stringing charts?
2	20-21 5(c)	10	numerical	A 132 KV transmission line has the following data: Weight of the conductor= 680 kg/km, Length of span= 260 m, Ultimate strength= 3100 kg, Safety Factor= 2. Calculate the height above ground at which the conductor should be supported. Ground clearance required is 10 meters.
Supports				
3	21-22 3(b)	5	theory	What kind of line supports are used for distribution purposes at low voltages (upto 11 kV)?

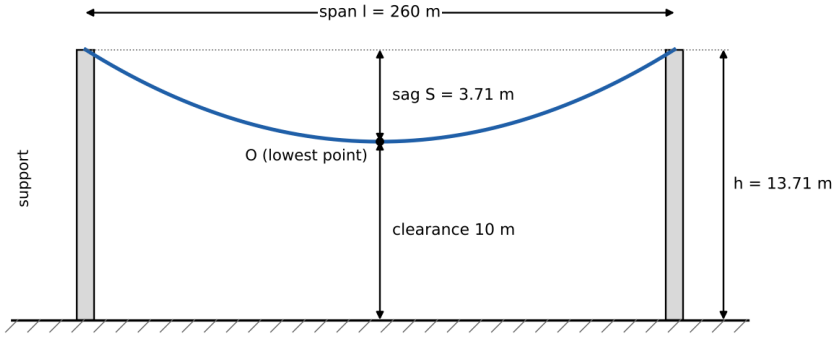


Figure 4: 20-21 5(c): supports at the same level, sag S and clearance

4.6. Underground cables:

85 marks, 11 questions, in 2 of 2 papers. Book: Mehta Ch. 11.

Most often numerical (8 of 11). Subtopics: Dielectric stress; Capacitance; Construction; Grading; Insulation resistance; Economical size.

Formulas:

Subtopic	Formula	Equation	Used in
Dielectric stress (43)	Flux density in the dielectric	$D_x = \frac{Q}{2\pi x}$	20-21 3(c)
	Stress from the charge	$g_x = \frac{Q}{2\pi\varepsilon_0\varepsilon_r\, x}$	20-21 3(c)
	Core-to-sheath voltage	$V = \frac{Q}{2\pi\varepsilon_0\varepsilon_r} \ln\frac{D}{d}$	20-21 3(c)
	Dielectric stress	$g_x = \frac{V}{x\,\ln(D/d)}$	20-21 3(c)
	Maximum and minimum stress	$g_{max} = \frac{2V}{d\,\ln(D/d)}$, $g_{min} = \frac{2V}{D\,\ln(D/d)}$	21-22 2(c), 20-21 2(d)

Subtopic	Formula	Equation	Used in
	Stress ratio	$\frac{g_{max}}{g_{min}} = \frac{D}{d}$	20-21 4(c)
	Insulation thickness	$t = \frac{D-d}{2}$	20-21 4(c)
	Operating voltage	$V = g_{max}\frac{d}{2}\ln\frac{D}{d}$	20-21 4(c)
	Cable capacitance	$C = \frac{2\pi\varepsilon_0\varepsilon_r}{\ln(D/d)}$ F/m	21-22 2(c), 20-21 3(c)
	Stress in the dielectric	$g = \frac{Q}{2\pi\varepsilon_0\varepsilon_r\, x} = \frac{V}{x\,\ln(D/d)}$	21-22 2(c)
	Cable charging current	$I_c = 2\pi f C V$	21-22 2(c)
Capacitance (10)	**Belted cable tests**	$C_1 = 3C_s,\ C_2 = C_s + 2C_c$	20-21 3(d)
	Capacitance per phase	$C_N = C_s + 3C_c$	20-21 3(d)
	Cable charging current	$I_c = 2\pi f\, C_N\, V_{ph}$	20-21 3(d)
Grading (20)	Layer stresses	$g_{1max} = \frac{Q}{\pi\varepsilon_0\varepsilon_1 d},\ g_{2max} = \frac{Q}{\pi\varepsilon_0\varepsilon_2 d_1}$	21-22 4(d)
	Two dielectrics, stresses	$\varepsilon_1 d\, g_{1max} = \varepsilon_2 d_1\, g_{2max}$	21-22 4(d)
	Two dielectrics, voltage	$V = \frac{g_{1max}\,d}{2}\ln\frac{d_1}{d} + \frac{g_{2max}\,d_1}{2}\ln\frac{D}{d_1}$	21-22 4(d)
	Capacitance grading	$V = \frac{g_{1max}\,d}{2}\ln\frac{d_1}{d} + \frac{g_{2max}\,d_1}{2}\ln\frac{d_2}{d_1} + \frac{g_{3max}\,d_2}{2}\ln\frac{D}{d_2}$	21-22 2(d)
	Equal maximum stress	$\varepsilon_1 d = \varepsilon_2 d_1 = \varepsilon_3 d_2$	21-22 2(d)
	Graded voltage	$V = \frac{g_{max}}{2}\left[d\ln\frac{d_1}{d} + d_1\ln\frac{d_2}{d_1} + d_2\ln\frac{D}{d_2}\right]$	21-22 2(d)
Insulation resistance (5)	**Insulation resistance**	$R = \frac{\rho}{2\pi l}\ln\frac{r_2}{r_1}$	21-22 4(b)
Economical size (2)	**Most economical size**	$d = \frac{2V}{g_{max}},\ D = e\,d = 2.718\,d$	21-22 6(a)

Symbols:

D_x = electric flux density at x ($\mathrm{C/m^2}$) ↩

Q = charge on the core (C/m) ↩

x = distance from the centre (cm) ↩

g_x = dielectric stress at x (V/m) ↩

ε_0 = permittivity of free space (8.854×10^{-12} F/m) ↩

ε_r = relative permittivity ↩

V = core-to-sheath voltage (kV) ↩

d = core diameter (cm) ↩

D = inside diameter of the sheath (cm) ↩

g_{max}, g_{min} = stress at the core and at the sheath (kV/cm) ↩

t = insulation thickness (m) ↩

C = capacitance (F/m) ↩

g = dielectric stress at x (kV/cm) ↩

I_c = charging current (A) ↩

f = frequency (Hz) ↩

C_1, C_2 = capacitances measured in tests 1 and 2 (F) ↩

C_s = core to sheath (F) ↩

C_c = core to core (F) ↩

C_N = capacitance per phase to

neutral (F) ↩

V_{ph} = phase voltage, $V_L/\sqrt{3}$ (V) ↩

g_{1max}, g_{2max} = largest stresses in layers 1 and 2 (kV/cm) ↩

ε_1, ε_2 = relative permittivities of the inner and outer layers ↩

d_1 = outer diameter of layer 1 (cm) ↩

g_{3max} = largest stress in layer 3 (kV/cm) ↩

d_2 = outer diameter of layer 2 (cm) ↩

ε_3 = relative permittivity of layer 3 ↩

R = insulation resistance (Ω) ↩

ρ = resistivity (Ω · cm) ↩

l = length (cm) ↩

r_1, r_2 = core and sheath radii (cm) ↩

e = 2.718, the base of natural logarithms ↩

Questions:

No.	Paper	Marks	Type	Question
Dielectric stress				
1	21-22 2(c)	8	numerical	A single core cable for use on 11 kV, 50 Hz system has conductor area of 0.645 cm^2 and internal diameter of sheath is 2.18 cm. The permittivity of the dielectric used in the cable is 3.5. Find: (i) the maximum electrostatic stress in the cable, (ii) minimum electrostatic stress in the cable, (iii) capacitance of the cable per km length, and (iv) charging current.
2	21-22 TT2 8(b)	5	numerical repeats 21-22 2(c)	A single core cable for use on 11 kV, 50 Hz system has a conductor area of 0.645 cm^2 and internal diameter of sheath is 2.18 cm. The permittivity of the dielectric used in the cable is 3.5. Find (i) the maximum electrostatic stress in the cable (ii) the minimum electrostatic stress in the cable (iii) the capacitance of the cable per km length (iv) the charging current.
3	20-21 2(d)	10	numerical	A 33 kV single core cable has a conductor diameter of 1 cm and a sheath of inside diameter 4 cm. Find the maximum and minimum stress in the insulation.
4	20-21 3(c)	10	derivation	Derive the expression of dielectric stress in a single core cable.
5	20-21 4(c)	10	numerical	The maximum and minimum stresses in the dielectric of a single core cable are 40 kV/cm (r.m.s.) and 10 kV/cm (r.m.s.) respectively. If the conductor diameter is 2 cm, find : (i) thickness of insulation (ii) operating voltage
Capacitance				
6	20-21 3(d)	10	numerical	The capacitances of a 3-phase belted cable are $12 \cdot 6$ μF between the three cores bunched together and the lead sheath and $7 \cdot 4$ μF between one core and the other two connected to sheath. Find the charging current drawn by the cable when connected to 66 kV, 50 Hz supply.
Construction				
7	21-22 2(b)	5	theory	Write down the general construction of a 3-conductor cable with required figure.
Grading				
8	21-22 2(d)	10	derivation	Define grading of cables. Why is unequal stress distribution in a cable undesirable? Consider three dielectric layers for capacitance grading and derive the expression of potential difference between core and earthed sheath.
9	21-22 4(d)	10	numerical	A 66 kV single-core lead sheathed cable is graded by using two dielectrics of relative permittivity 5 and 3 respectively; thickness of each being 1 cm. The core diameter is 2 cm. Determine the maximum stress in the two dielectrics.
Insulation resistance				
10	21-22 4(b)	5	numerical	The insulation resistance of a single-core cable is 495 MΩ per km. If the core diameter is 2.5 cm and the resistivity of insulation is 4.5×10^{14} $\Omega - cm$, find the insulation thickness.
Economical size				
11	21-22 6(a)	2	numerical	Find the most economical size of a single-core cable working on a 132 kV, 3-phase system, if a dielectric stress of 60 kV/cm can be allowed.

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4.7. Cable faults:

20 marks, 3 questions, in 2 of 2 papers. Book: Mehta Ch. 11.

Most often numerical (3 of 3). Subtopics: Loop tests.

Formulas:

Subtopic	Formula	Equation	Used in
Loop tests (20)	**Varley loop balance**	$\frac{P}{Q} = \frac{R_{loop} - R_X}{R_X + S}$, $R_X = \frac{Q\,R_{loop} - P\,S}{P+Q}$	20-21 5(b)
	Equal ratio arms	$R_X = \frac{R_{loop} - S}{2}$	20-21 5(b)
	Murray loop balance	$\frac{P}{Q} = \frac{R_{loop} - R_X}{R_X}$, $R_X = \frac{Q}{P+Q} R_{loop}$	21-22 1(b)

Symbols:

P, Q = ratio arms (Ω) ↩

R_X = resistance of the faulty conductor up to the fault (Ω) ↩

S = variable resistance in series with the faulty conductor (Ω) ↩

R_{loop} = faulty + healthy conductor (Ω); faulty + sound core (Ω) ↩

Questions:

No.	Paper	Marks	Type	Question
Loop tests				
1	21-22 1(b)	5	numerical	In a test for a fault to earth on a 500 m length of cable having a resistance of 1 Ω per 1000 m, the faulty cable is looped with a sound cable of the same length but having a resistance of 2.25 Ω per 1000 m. The resistance of the other two arms of the testing network at balance are in the ratio of 2.75 : 1. Calculate the distance of the fault from the testing end of the cable.
2	21-22 TT3 9(a)	5	numerical repeats 21-22 1(b)	In a test for a fault to earth on a 500 m length of cable having a resistance of 1 Ω per 1000 m, the faulty cable is looped with a sound cable of the same length but having a resistance of 2.25 Ω per 1000 m. The resistance of the other two arms of the testing network at balance are in the ratio 2.75 : 1. Calculate the distance of the fault from the testing end of the cable.
3	20-21 5(b)	10	numerical	Varley loop test is performed to locate an earth fault on a 20 km long cable. The resistance per km of the single conductor is 20 ohm. The loop is completed with a similar healthy conductor. At balance, the variable resistance connected to the faulty conductor is 200 ohm. The fixed resistors have equal values. Calculate the distance of the fault from the test end.

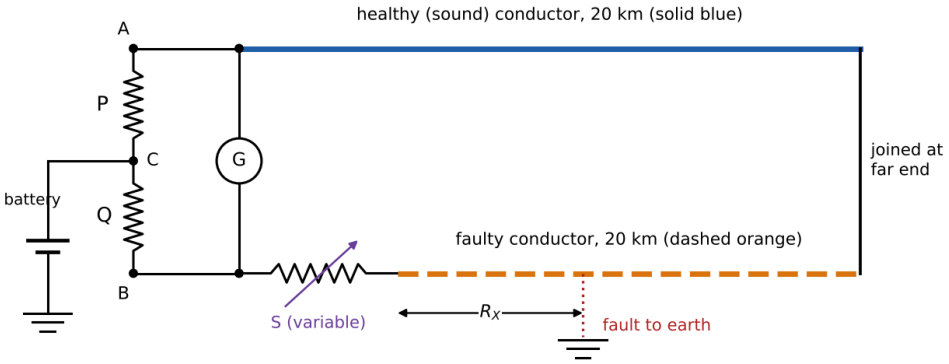


Figure 5: 20-21 5(b): the Varley loop with S in series with the faulty conductor

4.8. Power system stability:

43 marks, 5 questions, in 2 of 2 papers. Book: Grainger Ch. 16.

Most often theory (3 of 5). Subtopics: Equal area criterion; Transient EMF.

Formulas:

Subtopic	Formula	Equation	Used in
Equal area criterion (23)	**Power-angle equation**	$P_e = \frac{\lvert E\rvert\lvert V\rvert}{X}\sin\delta = P_{max}\sin\delta$	20-21 6(a)
	Swing equation	$M\frac{d^2\delta}{dt^2} = P_m - P_e$	21-22 6(c), 20-21 6(a)
	Equal area criterion	$\int_{\delta_0}^{\delta_{max}} (P_m - P_e)\,d\delta = 0 \;\Leftrightarrow\; A_1 = A_2$	20-21 6(a)
	Critical clearing angle	$\cos\delta_{cr} = \frac{P_m(\delta_{max}-\delta_0) + P_{3max}\cos\delta_{max} - P_{2max}\cos\delta_0}{P_{3max} - P_{2max}}$	20-21 6(a)
	Limit of the swing	$\delta_{max} = 180° - \delta_1$	20-21 6(b)
	Equal area criterion	$A_1 = A_2$	21-22 6(c)
Transient EMF (20)	Terminal voltage angle	$\sin\theta = \frac{P\,X}{\lvert V_t\rvert\lvert V\rvert}$	20-21 6(c)
	Current	$I = \frac{V_t - V}{jX}$	20-21 6(c)
	Transient emf	$E' = V_t + jX'_dI$	21-22 6(d), 20-21 6(c)
	Maximum power	$P_{max} = \frac{\lvert E'\rvert\lvert V\rvert}{X}$	20-21 6(c)
	Transfer reactance (Y to Δ)	$X_{ab} = \frac{X_aX_b + X_bX_c + X_cX_a}{X_c}$	21-22 6(d), 20-21 6(c)
	Power-angle equation	$P_e = \frac{\lvert E'\rvert\lvert V\rvert}{X}\sin\delta = P_{max}\sin\delta$	21-22 6(d)

Symbols:

P_e = electrical power output (pu) ↩

E = internal emf of the machine (pu) ↩

V = infinite-bus voltage (pu) ↩

δ = power angle (E ahead of V); rotor angle ↩

P_{max} = peak of the curve, at $\delta = 90°$ (pu) ↩

M = inertia constant ↩

t = time (s)

P_m = mechanical input (pu) ↩

δ_0 = angle before the fault ↩

δ_{max} = largest angle of the swing ↩

A_1 = accelerating area

A_2 = decelerating area

δ_{cr} = critical clearing angle ↩

P_{2max}, P_{3max} = peaks of the power curve during and after the fault (pu) ↩

δ_1 = new operating angle ↩

A_1, A_2 = accelerating and decelerating areas ↩

θ = angle of V_t ahead of V ↩

P = power delivered (pu) ↩

X = reactance from V_t to the bus (pu); transfer reactance (pu) ↩

V_t = terminal voltage (pu) ↩

I = current (pu) ↩

E' = transient internal emf (pu) ↩

X'_d = transient reactance (pu) ↩

X_{ab} = transfer reactance between nodes a and b (pu) ↩

X_a, X_b = branches of the Y to nodes a and b (pu) ↩

X_c = branch to the faulted node (pu) ↩

Questions:

No.	Paper	Marks	Type	Question
Equal area criterion				
1	21-22 6(c)	8	theory repeats 20-21 6(b)	Explain what happens for sudden increase in mechanical input to generator using $Pe - \delta$ diagram.
2	20-21 6(a)	5	theory	Define Power Angle. Briefly explain Equal Area Criterion.
3	20-21 6(b)	10	theory	Explain what happens for sudden increase in mechanical input to generator using Pe-δ diagram.
Transient EMF				
4	21-22 6(d)	10	numerical repeats 20-21 6(c)	The generator of Fig. 2 is delivering 1.0 pu power to the infinite bus ($\|V\| = 1.0\ pu$), with the generator terminal voltage ($\|V_t\| = 1.0\ pu$). Calculate the generator emf behind transient reactance. Find the maximum power that can be transferred under the following conditions: (i) System healthy, (ii) One line shorted (3 phase) in the middle and (iii) One line open. Plot all the three power angle curves.
5	20-21 6(c)	10	numerical	The generator of Fig. 2 is delivering 1.0 pu power to the infinite bus (|V|=1.0 pu), with the generator terminal voltage (|Vt|=1.0 pu). Calculate the generator emf behind transient reactance. Find the maximum power that can be transferred under the following conditions: (a) System healthy (b) One line shorted (3 phase) in the middle.

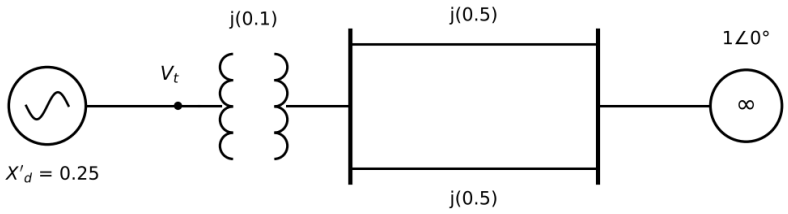


Figure 6: 21-22 6(d): Fig. 2 of the paper (redrawn): figure for Q-6(d)

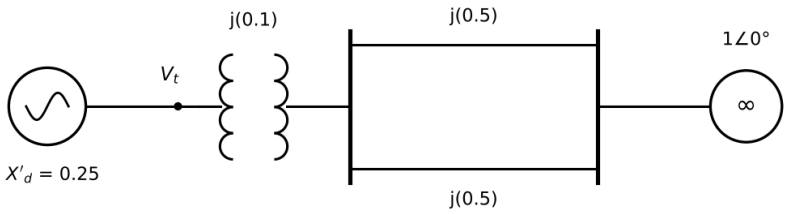


Figure 7: 20-21 6(c): Fig. 2 of the paper (redrawn)

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4.9. HVDC and FACTS:

11 marks, 3 questions, in 2 of 2 papers. Book: Hingorani and Gyugyi; Rashid Ch. 12.

Most often theory (3 of 3). Subtopics: FACTS; HVDC.

Formulas:

Subtopic	Formula	Equation	Used in
FACTS (4)	**Power over a line**	$P = \dfrac{V_1 V_2}{X} \sin\delta$	20-21 3(b)
HVDC (7)	Annual cost	$P_1 + P_2 a + \dfrac{P_3}{a}$	21-22 6(b)
	Kelvin's law	$P_2 a = \dfrac{P_3}{a}$	21-22 6(b)

Symbols:

P = power sent over the line (W) ↩ X = line reactance (Ω) ↩ a = conductor area ↩

V_1, V_2 = voltages at the two ends (V) ↩ δ = angle between V_1 and V_2 ↩ $P_2 a$ = interest and depreciation ↩

P_1 = fixed part of the annual cost ↩ P_3/a = cost of energy lost

Questions:

No.	Paper	Marks	Type	Question
FACTS				
1	20-21 3(b)	4	theory	Describe different FACTS devices and applications.
HVDC				
2	21-22 2(a)	2	theory	What is monopolar link?
3	21-22 6(b)	5	theory	Define HVDC. Discuss the Economics of Power transmission.

5. Theory points:

What a full answer covers, for every theory question, in book order. The link opens the full answer.

No.	Paper	Marks	Subtopic	Points to cover
Line inductance				
1	20-21 1(a)	1	Basics	Inductance is flux linkage per ampere, $L = \psi/I = N\phi/I$ henry; it opposes a change of current, $e = L\,di/dt$; for a line, $L = 2 \times 10^{-7} \ln(D/r')$ H/m per conductor.
2	20-21 2(b)	4	Transposition	A transposed line exchanges conductor positions at intervals so each occupies each position for one third of the length. Reason: with unsymmetrical spacing the phases have unequal L and C; transposition equalises them, keeps the voltage drops balanced and reduces telephone interference.
3	20-21 3(a)	1	Skin effect	Skin effect: alternating current crowds towards the surface because the inner filaments link more flux and have more reactance; the effective area falls, so the ac resistance exceeds the dc resistance; it grows with frequency, size and conductivity; stranded and hollow conductors reduce it.
Line capacitance				
4	21-22 5(a)	2	Effect of earth	The earth is a zero-potential conductor; by the method of images it acts as equal and opposite charges at the same depth below ground; this raises the line capacitance slightly ($C = \pi\varepsilon_0 / \ln[(d/r) \cdot 2h/\sqrt{4h^2 + d^2}]$ for a single-phase line); small for high lines.

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No.	Paper	Marks	Subtopic	Points to cover
Corona				
5	21-22 1(a)	2	Critical disruptive voltage	V_c: the least phase-to-neutral voltage at which the surface field reaches the breakdown strength of air (30 kV/cm peak) and corona starts, $V_c = m_0 g_0 \delta r \ln(d/r)$ kV; below it no corona; above it corona loss, interference and ozone; lines are designed below it.
6	20-21 2(a)	1	Corona effect	Corona: violet glow, hissing, ozone and power loss round a conductor when the surface field exceeds the strength of air (30 kV/cm peak) and the air ionises. Depends on voltage, conductor size and surface, spacing and weather.
7	20-21 4(a)	5	Critical disruptive voltage	V_c: the least phase-to-neutral voltage at which the surface field reaches the strength of air (30 kV/cm peak) and corona starts: $V_c = m_0 g_0 \delta r \ln(d/r)$ kV. Significance: above it corona loss, interference and ozone appear; lines are designed to work below it.
Overhead insulators				
8	21-22 3(a)	2	Failure	By flash-over (an arc through the air from the conductor to the pin; the insulator survives) or puncture (a discharge through the porcelain; destroyed). Safety factor = puncture strength / flash-over voltage, about 10, so flash-over comes first.
9	20-21 4(b)	10	String efficiency	$\eta = V/(nV_n)$; the shunt capacitance KC makes the line unit take the most voltage, so $\eta < 100$ %. Improve it with longer cross-arms (smaller K), capacitance grading of the discs, and a guard (static shielding) ring at line potential.
10	20-21 5(a)	5	Failure	Cracking from unequal expansion of porcelain, steel and cement; impure material; porosity; poor glazing; flash-over (arc heat cracks it); mechanical stress (tension, wind, ice); short circuits.
Mechanical design				
11	21-22 3(b)	5	Supports	Wooden poles (cheap, spans of 40 to 50 m, rural), steel poles (rail or tubular, 50 to 80 m, towns), RCC poles (strong, long life, 80 to 200 m); A- or H-type double poles for heavy loads.
12	21-22 4(a)	2	Sag	A stringing chart: curves of sag and tension against temperature for a given conductor and span; the lineman reads the right sag (or tension) for the temperature at erection, so the line is safe at both temperature extremes.
Underground cables				
13	21-22 2(b)	5	Construction	From the centre: three stranded cores, each paper (or PVC/XLPE) insulated; jute fillers; a common paper belt; lead sheath; jute bedding; steel-wire or tape armour; jute serving.
Power system stability				
14	21-22 6(c)	8	Equal area criterion	A sudden rise of P_m makes $P_m > P_e$: the rotor accelerates, δ passes the new equilibrium δ_1 (area A_1), overshoots to δ_2 where the decelerating area $A_2 = A_1$, then swings back and, with damping, settles at δ_1; unstable if A_1 exceeds the area available up to $180° - \delta_1$.
15	20-21 6(a)	5	Equal area criterion	Power angle δ: the angle between the internal emf and the infinite-bus voltage; $P_e = (EV/X)\sin\delta$. Equal area criterion: after a disturbance the machine stays in step if the decelerating area ($P_e > P_m$) can equal the accelerating area ($P_m > P_e$).
16	20-21 6(b)	10	Equal area criterion	A sudden rise of P_m makes $P_m > P_e$: the rotor accelerates, δ passes the new equilibrium δ_1 (area A_1), overshoots to δ_2 where the decelerating area $A_2 = A_1$, then swings back and, with damping, settles at δ_1; unstable if A_1 exceeds the area available up to $180° - \delta_1$.
HVDC and FACTS				
17	21-22 2(a)	2	HVDC	Monopolar HVDC link: one conductor (usually negative) between the converter stations, earth or sea as the return; cheap, used for submarine cables; bipolar and homopolar are the other kinds.

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No.	Paper	Marks	Subtopic	Points to cover
18	21-22 6(b)	5	HVDC	HVDC: power as high-voltage dc between a rectifier and an inverter station. Economics: Kelvin's law for conductor size (annual loss cost = annual capital charge on the area-dependent part), an economic voltage, and HVDC cheaper than HVAC beyond the break-even distance (lines and cables) despite costly terminals.
19	20-21 3(b)	4	FACTS	FACTS controllers change V, X or δ in $P = V_1V_2 \sin\delta/X$. Shunt (SVC, STATCOM): voltage control and reactive support. Series (TCSC, SSSC): lower X, more power, flow control, damping. Combined (UPFC, IPFC): V, X and δ together.

6. Derivations:

Every derivation asked, in book order. Each outline gives the steps; the link opens the full working.

No.	Paper	Marks	Derivation	Outline
Line inductance				
1	21-22 1(c)	8	Transposition	Flux of a line current $2\times10^{-7}I\ln(1/D)$; ψ_a in the three sections; average: the distances multiply; $I_b + I_c = -I_a$; $\psi_a = 2\times10^{-7}I_a\ln(D_{eq}/r')$; $L = \psi_a/I_a$.
2	21-22 4(c)	8	Composite conductors	Flux of a line current $2\times10^{-7}I\ln(1/D)$, with r' for itself; flux of strand a; combine the logs; $L_a = \psi_a/(I/n)$; every strand; n strands in parallel, $L_X = \sum L/n^2$; mutual GMD over self GMD; loop $L = L_X + L_Y$.
3	20-21 2(c)	10	Composite conductors	Flux of a line current $2\times10^{-7}I\ln(1/D)$, with r' for itself; flux of strand a; combine the logs; $L_a = \psi_a/(I/n)$; every strand; n strands in parallel, $L_X = \sum L/n^2$; mutual GMD over self GMD; loop $L = L_X + L_Y$.
Line capacitance				
4	21-22 5(d)	10	Unsymmetrical spacing	Line charge: $V = (q/2\pi\varepsilon_0)\ln(D_2/D_1)$; V_{ab} in the three transposition sections; average: the distances multiply, $D_{eq} = \sqrt[3]{D_{12}D_{23}D_{31}}$; same for V_{ac}; add with $q_a + q_b + q_c = 0$; $V_{ab} + V_{ac} = 3V_{an}$; $C_n = 2\pi\varepsilon_0/\ln(D_{eq}/r)$.
5	21-22 TT1 7(b)	0	Unsymmetrical spacing	Line charge: $V = (q/2\pi\varepsilon_0)\ln(D_2/D_1)$; V_{ab} in the three transposition sections; average: the distances multiply, $D_{eq} = \sqrt[3]{D_{12}D_{23}D_{31}}$; same for V_{ac}; add with $q_a + q_b + q_c = 0$; $V_{ab} + V_{ac} = 3V_{an}$; $C_n = 2\pi\varepsilon_0/\ln(D_{eq}/r)$.
6	20-21 1(c)	10	Unsymmetrical spacing	Line charge: $V = (q/2\pi\varepsilon_0)\ln(D_2/D_1)$; V_{ab} in the three transposition sections; average: the distances multiply, $D_{eq} = \sqrt[3]{D_{12}D_{23}D_{31}}$; same for V_{ac}; add with $q_a + q_b + q_c = 0$; $V_{ab} + V_{ac} = 3V_{an}$; $C_n = 2\pi\varepsilon_0/\ln(D_{eq}/r)$.
Overhead insulators				
7	21-22 3(d)	10	String efficiency	Each capacitance carries ωCV; KCL at A: $V_2 = (1+K)V_1$; KCL at B: $V_3 = (1+3K+K^2)V_1$; add for V; $\eta = V/(3V_3)$; evaluate at $K = 0.2$, 0.8 and for more discs.
Underground cables				
8	21-22 2(d)	10	Grading	Gauss: $D_x = Q/2\pi x$ in every layer; $g_k = Q/(2\pi\varepsilon_0\varepsilon_k x)$; maximum at each inner surface; integrate each layer; add the three; equal maxima need $\varepsilon_1 d = \varepsilon_2 d_1 = \varepsilon_3 d_2$.
9	20-21 3(c)	10	Dielectric stress	Gauss: $D_x = Q/2\pi x$; $g_x = Q/(2\pi\varepsilon_0\varepsilon_r x)$; integrate from $d/2$ to $D/2$: $V = (Q/2\pi\varepsilon_0\varepsilon_r)\ln(D/d)$; remove Q: $g_x = V/(x\ln(D/d))$; g_{max} at the core, g_{min} at the sheath.

7. Numericals:

Every numerical asked, in book order: the method in one line and the final answer, so a worked attempt can be checked. The link opens the full working.

No.	Paper	Marks	Subtopic	Method	Answer
Line inductance					
1	21-22 1(d)	10	Mutual inductance	D_1 and D_2 from the geometry of Fig. 1; $M = 4 \times 10^{-7} \ln(D_2/D_1)$ (both power wires add); $E = \omega M I$.	$M = 0.0758$ mH/km; 2.38 V/km
2	21-22 TT1 7(a)	5	Bundled conductors	$r' = 0.7788r$; bundle $D_s = \sqrt{r'd}$; D_{eq} from the bundle centres; $L = 2 \times 10^{-7} \ln(D_{eq}/D_s)$; $X_L = 2\pi f L$.	$L = 1.102$ mH/km; $X_L = 0.346$ Ω/km
3	21-22 TT3 9(b)	5	Mutual inductance	Distances from each power wire to each telephone wire; flux of P_1 and P_2 linking the telephone loop; $M = 2 \times 10^{-7} \ln[(D_{P_1T_2} D_{P_2T_1})/(D_{P_1T_1} D_{P_2T_2})]$; $E = \omega M I$.	$M = 0.648$ μH/km; 30.6 mV/km
4	20-21 1(b)	4	Bundled conductors	Bundle GMR $D_s^b = \sqrt{D_s d}$ with the table D_s; $D_{eq} = \sqrt[3]{D_{ab} D_{bc} D_{ca}}$; $L = 2 \times 10^{-7} \ln(D_{eq}/D_s^b)$; $X_L = 2\pi f L$ at 50 Hz (handwritten on the scan).	$X_L = 0.304$ Ω/km = 0.489 Ω/mi per phase (0.365 Ω/km at 60 Hz)
5	20-21 1(d)	10	Mutual inductance	Distances from each power wire to each telephone wire; flux of P_1 and P_2 linking the telephone loop; $M = 2 \times 10^{-7} \ln[(D_{P_1T_2} D_{P_2T_1})/(D_{P_1T_1} D_{P_2T_2})]$; $E = \omega M I$.	$M = 0.648$ μH/km; 30.6 mV/km
Line capacitance					
6	21-22 5(b)	5	Unsymmetrical spacing	$D_{eq} = \sqrt[3]{D \cdot D \cdot 2D}$; $C_n = 2\pi\varepsilon_0 / \ln(D_{eq}/r)$; $I_c = \omega C_n V_{ph}$.	$C_n = 0.00826$ μF/km; 0.165 A/km
Overhead insulators					
7	21-22 3(c)	8	String efficiency	$V = 33/\sqrt{3}$ kV; $V_2 = (1+K)V_1$, $V_3 = (1+3K+K^2)V_1$; V_1 from $V = V_1 + V_2 + V_3$; $\eta = V/(3V_3)$.	5.52, 6.13, 7.41 kV; $\eta = 85.74$ %
8	21-22 5(c)	8	String efficiency	$V_2 = (1+K)V_1$ gives K; $V_3 = V_2 + K(V_1 + V_2)$; phase voltage $= V_1 + V_2 + V_3$; line voltage $\sqrt{3}$ times; $\eta = V/(3V_3)$.	$K = 0.375$; $V_L = 64.3$ kV; $\eta = 68.28$ %
9	21-22 TT2 8(a)	5	String efficiency	$V = 100/\sqrt{3}$ kV; $V_{k+1} = V_k + K(V_1 + \cdots + V_k)$ for five discs; V_1 from the sum; $\eta = V/(5V_5)$.	8.00, 8.80, 10.48, 13.21, 17.25 kV; $\eta = 66.92$ %
Mechanical design					
10	20-21 5(c)	10	Sag	Working tension T = ultimate strength / safety factor; sag $S = wl^2/(8T)$; support height = clearance + sag.	sag 3.71 m; supports at 13.71 m
Underground cables					
11	21-22 2(c)	8	Dielectric stress	$g_{max} = 2V/(d \ln(D/d))$ at the core, $g_{min} = 2V/(D \ln(D/d))$ at the sheath; $C = 2\pi\varepsilon_0\varepsilon_r / \ln(D/d)$; $I_c = \omega C V$.	$g_{max} = 27.66$, $g_{min} = 11.50$ kV/cm; $C = 0.2218$ μF/km; $I_c = 0.767$ A/km
12	21-22 4(b)	5	Insulation resistance	$R = (\rho/2\pi l) \ln(r_2/r_1)$; solve $\ln(r_2/r_1) = 2\pi l R/\rho$; thickness $r_2 - r_1$.	1.245 cm
13	21-22 4(d)	10	Grading	The same Q passes both layers, so $g_{2max}/g_{1max} = \varepsilon_1 d/(\varepsilon_2 d_1)$; $V = (g_{1max} d/2) \ln(d_1/d) + (g_{2max} d_1/2) \ln(D/d_1)$; solve for g_{1max}.	48.21 kV/cm (inner, $\varepsilon_r = 5$); 40.18 kV/cm (outer) r.m.s.
14	21-22 6(a)	2	Economical size	Peak phase voltage $V = 132\sqrt{2}/\sqrt{3}$; most economical core $d = 2V/g_{max}$ (from $D/d = e$); sheath $D = ed$.	$d = 3.59$ cm; $D = 9.77$ cm

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No.	Paper	Marks	Subtopic	Method	Answer
15	21-22 TT2 8(b)	5	Dielectric stress	$g_{max} = 2V/(d\ln(D/d))$ at the core, $g_{min} = 2V/(D\ln(D/d))$ at the sheath; $C = 2\pi\varepsilon_0\varepsilon_r/\ln(D/d)$; $I_c = \omega CV$.	$g_{max} = 27.66$, $g_{min} = 11.50$ kV/cm; $C = 0.2218$ μF/km; $I_c = 0.767$ A/km
16	20-21 2(d)	10	Dielectric stress	$g_{max} = 2V/(d\ln(D/d))$, $g_{min} = 2V/(D\ln(D/d))$ with $V = 33$ kV core to sheath.	$g_{max} = 47.61$ kV/cm, $g_{min} = 11.90$ kV/cm (r.m.s.)
17	20-21 3(d)	10	Capacitance	Three cores bunched to sheath: $3C_s$ measured; one core to the other two and sheath: $C_s + 2C_c$; per-phase $C_N = C_s + 3C_c$; $I_c = \omega C_N V_{ph}$.	$C_s = 4.2$, $C_c = 1.6$, $C_N = 9.0$ μF; $I_c = 107.7$ A
18	20-21 4(c)	10	Dielectric stress	$g_{max}/g_{min} = D/d$ gives D; thickness $(D-d)/2$; $V = g_{max}(d/2)\ln(D/d)$.	thickness 3 cm ($D = 8$ cm); $V = 55.45$ kV r.m.s.
Cable faults					
19	21-22 1(b)	5	Loop tests	Loop resistance R_{loop} = faulty + sound cable; at balance $R_X/(R_{loop} - R_X) = Q/P$, so $R_X = R_{loop}/(1 + P/Q)$; distance = R_X / resistance per metre.	fault at 433.3 m from the testing end
20	21-22 TT3 9(a)	5	Loop tests	Loop resistance R_{loop} = faulty + sound cable; at balance $R_X/(R_{loop} - R_X) = Q/P$, so $R_X = R_{loop}/(1 + P/Q)$; distance = R_X / resistance per metre.	fault at 433.3 m from the testing end
21	20-21 5(b)	10	Loop tests	Varley loop: R_{loop} = twice the cable resistance; with $P = Q$, $R_X = (R_{loop} - S)/2$; distance = R_X / resistance per km.	15 km from the test end
Power system stability					
22	21-22 6(d)	10	Transient EMF	V_t angle from $P = V_t V \sin\theta/X$; current $I = (V_t - V)/jX$; $E' = V_t + jX'_dI$; $P_{max} = E'V/X_{transfer}$ for each network (fault: star-delta or Thevenin reduction).	$E' = 1.0747\angle 33.94°$ pu; $P_{max} = 1.791$ (healthy), 0.693 (mid-line fault), 1.264 pu (one line open)
23	20-21 6(c)	10	Transient EMF	V_t angle from $P = V_t V \sin\theta/X$; current $I = (V_t - V)/jX$; $E' = V_t + jX'_dI$; $P_{max} = E'V/X_{transfer}$ for each network (fault: star-delta or Thevenin reduction).	$E' = 1.0747\angle 33.94°$ pu; $P_{max} = 1.791$ pu healthy, 0.693 pu with a mid-line fault

8. Diagrams to practise:

Every figure an answer needs, whatever the question's type, in book order. Draw each one from memory, then check it against the solution.

No.	Paper	Kind	Diagram
Line inductance			
1	21-22 1(c)	Diagram	flux of one conductor
2	21-22 1(c)	Diagram	one transposition cycle
3	21-22 1(d)	Diagram	flux of the power wires linking the telephone loop
4	21-22 4(c)	Diagram	composite conductors X (n strands) and Y (m strands)
5	21-22 TT1 7(a)	Diagram	two-conductor bundle and its GMR
6	21-22 TT3 9(b)	Diagram	power and telephone lines in one plane
7	20-21 1(a)	Diagram	flux of one current-carrying conductor, inside and outside

No.	Paper	Kind	Diagram
8	20-21 1(b)	Diagram	two-conductor bundle and its GMR
9	20-21 1(d)	Diagram	power and telephone lines in one plane
10	20-21 2(b)	Diagram	one transposition cycle
11	20-21 2(c)	Diagram	composite conductors X (n strands) and Y (m strands)
12	20-21 3(a)	Diagram	current density across a conductor, dc and ac
Line capacitance			
13	21-22 5(a)	Diagram	two-wire line above the earth with image charges
14	21-22 5(b)	Diagram	three conductors in a horizontal row
15	21-22 5(d)	Diagram	three charged conductors at unequal spacing
16	21-22 5(d)	Diagram	one transposition cycle
17	21-22 TT1 7(b)	Diagram	three charged conductors at unequal spacing
18	21-22 TT1 7(b)	Diagram	one transposition cycle
19	20-21 1(c)	Diagram	three charged conductors at unequal spacing
20	20-21 1(c)	Diagram	one transposition cycle
Corona			
21	21-22 1(a)	Diagram	field lines and the corona shell round a conductor
22	20-21 2(a)	Diagram	field lines and the ionised shell round a conductor
Overhead insulators			
23	21-22 3(a)	Diagram	flash-over and puncture paths of a pin insulator
24	21-22 3(c)	Circuit	equivalent circuit of a three-disc string
25	21-22 3(d)	Graph	voltage share per disc for several K
26	21-22 3(d)	Graph	efficiency against the number of discs
27	21-22 5(c)	Circuit	equivalent circuit of the three-disc string with the measured voltages
28	21-22 TT2 8(a)	Graph	voltage on each disc of the five-disc string
29	20-21 4(b)	Circuit	equivalent circuit of a three-disc string
30	20-21 4(b)	Diagram	three ways of improving string efficiency
31	20-21 4(b)	Graph	efficiency against the number of discs
32	20-21 5(a)	Diagram	flash-over and puncture paths of a pin insulator
Mechanical design			
33	21-22 3(b)	Diagram	wooden, steel and RCC poles
34	21-22 4(a)	Graph	sag and tension against temperature
35	20-21 5(c)	Diagram	conductor between supports with sag and clearance
Underground cables			
36	21-22 2(b)	Diagram	cross-section of a three-core belted cable
37	21-22 2(c)	Graph	stress across the insulation, largest at the core
38	21-22 2(d)	Diagram	cable with three dielectric layers and the stress profile
39	21-22 4(b)	Diagram	radial leakage current in the insulation
40	21-22 4(d)	Graph	stress across the graded and ungraded cable
41	21-22 6(a)	Graph	sheath diameter against core diameter
42	21-22 TT2 8(b)	Graph	stress across the insulation, largest at the core
43	20-21 2(d)	Graph	stress profile of this cable
44	20-21 3(c)	Diagram	cable section with the Gauss cylinder
45	20-21 3(d)	Diagram	belted cable construction
46	20-21 3(d)	Diagram	core-to-core and core-to-sheath capacitances
47	20-21 4(c)	Graph	stress across this cable

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No.	Paper	Kind	Diagram
Cable faults			
48	21-22 1(b)	Circuit	Murray loop test
49	21-22 TT3 9(a)	Circuit	Murray loop test
50	20-21 5(b)	Circuit	Varley loop test
Power system stability			
51	21-22 6(c)	Graph	P_e–δ curve with the accelerating and decelerating areas
52	21-22 6(c)	Graph	rotor angle swing after the step
53	21-22 6(d)	Diagram	network with the mid-line fault and its reduction
54	21-22 6(d)	Graph	power-angle curves for the three cases
55	20-21 6(a)	Diagram	phasor diagram with the power angle
56	20-21 6(a)	Graph	equal area criterion for a cleared fault
57	20-21 6(b)	Graph	P_e–δ curve with the accelerating and decelerating areas
58	20-21 6(b)	Graph	rotor angle swing after the step
59	20-21 6(c)	Diagram	network with the mid-line fault and its reduction
60	20-21 6(c)	Graph	power-angle curves
HVDC and FACTS			
61	21-22 2(a)	Diagram	monopolar, bipolar and homopolar HVDC links
62	21-22 6(b)	Graph	Kelvin's law, capital charge and loss cost against area
63	21-22 6(b)	Graph	total cost against distance for HVAC and HVDC
64	20-21 3(b)	Diagram	shunt, series and combined FACTS connections

9. Comparisons:

Every comparison asked, in book order, with the key difference in one line. A pair asked in several papers is one row.

Pair	Key difference	Asked in
Corona		
Corona vs skin effect	air ionises round a high-voltage conductor vs ac current crowds to the conductor surface	20-21 2(a)
Critical disruptive vs visual critical voltage	corona starts (no glow yet) vs the glow becomes visible, higher	20-21 4(a)
Overhead insulators		
Flash-over vs puncture	arc through the air outside, insulator survives vs discharge through the porcelain, insulator destroyed	21-22 3(a), 20-21 5(a)
Underground cables		
Capacitance vs intersheath grading	layers of different permittivity vs metal intersheaths held at set voltages	21-22 2(d)
Cable faults		
Murray vs Varley loop test	ratio arms only, balance by P/Q vs extra variable resistor S in the faulty leg	20-21 5(b)
HVDC and FACTS		
SVC vs STATCOM	thyristor-switched reactors and capacitors vs voltage-source converter, works at low voltage	20-21 3(b)
Monopolar vs bipolar vs homopolar HVDC link	one conductor, earth return vs two conductors of opposite polarity vs two conductors of the same polarity, earth return	21-22 2(a)
HVDC vs HVAC transmission	costly terminals, cheap lines, no stability limit vs cheap terminals, reactive power, stability limit	21-22 6(b)

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10. Repeated questions:

8 questions came back, exactly or with new numbers; 1 of them in three or more papers. Most repeated first; the latest paper first in each row.

No.	Times	Topic	Subtopic	Asked in (marks)
1	3	Line capacitance	Unsymmetrical spacing	21-22 5(d) (10), 21-22 TT1 7(b), 20-21 1(c) (10)
2	2	Line inductance	Mutual inductance	21-22 TT3 9(b) (5), 20-21 1(d) (10)
3	2	Line inductance	Composite conductors	21-22 4(c) (8), 20-21 2(c) (10)
4	2	Corona	Critical disruptive voltage	21-22 1(a) (2), 20-21 4(a) (5)
5	2	Power system stability	Equal area criterion	21-22 6(c) (8), 20-21 6(b) (10)
6	2	Power system stability	Transient EMF	21-22 6(d) (10), 20-21 6(c) (10)
7	2	Cable faults	Loop tests	21-22 1(b) (5), 21-22 TT3 9(a) (5)
8	2	Underground cables	Dielectric stress	21-22 2(c) (8), 21-22 TT2 8(b) (5)

A sub-part with no marks after it had none printed on its paper.

11. Question style:

How each topic was asked, as a number of questions.

Topic	Theory	Derivation	Numerical
Line inductance	3	3	5
Line capacitance	1	3	1
Corona	3	·	·
Overhead insulators	3	1	3
Mechanical design	2	·	1
Underground cables	1	2	8
Cable faults	·	·	3
Power system stability	3	·	2
HVDC and FACTS	3	·	·

12. Study order:

Topics by marks offered, with the share of all marks covered so far. Papers offer more marks than you answer, so this is a guide to where marks have been, not a score.

No.	Topic	Marks	Covered so far
1	Underground cables	85	26 %
2	Line inductance	66	46 %
3	Overhead insulators	48	61 %
4	Power system stability	43	74 %
5	Line capacitance	27	83 %
6	Cable faults	20	89 %
7	Mechanical design	17	94 %
8	HVDC and FACTS	11	98 %
9	Corona	8	100 %

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